

Research Assignment - Module 01

The Language of Business Analytics

Section 1: Foundational Concepts of Analytics

1. **Business Analytics:** When companies use data to understand what's going on in the business and make better decisions. It's not just looking at numbers, but actually using them to improve performance and strategy.

Example: Shoprite uses sales data to decide which products to stock more of in different areas.

Source: Investopedia (2024)

2. **Data Analytics:** The process of analysing raw data to find useful insights. It focuses more on the data itself, while business analytics focuses on decision-making.

Example: Looking at how many people buy airline on different days to see when sales peak.

Source: IBM (2024) ~~Data Analytics Guide~~

3. Business Intelligence: Tools and systems that collect and present data in an easy way, like dashboards or reports. It mainly helps track what is happening in the business right now or in the past.

Example: A dashboard showing daily sales across Pick n Pay stores.

Source: Microsoft BI Documentation

4. Data - Driven Decision Making: When decisions are made based on data instead of just gut feel or assumptions. It helps businesses reduce risk and be more accurate.

Example: A company deciding to open a new store in Soneto after seeing high customer demand data from that area.

Source: Harvard Business Review

5. Big Data: Big Data refers to very large amounts of data that come in quickly and from different sources. It is often too complex to handle using normal tools.

Example: Data collected from millions of users on apps like WhatsApp or TikTok everyday.

Source: Oracle (2024)

6. Descriptive Analytics: Looking at past data to explain what has already happened.

Example: A ~~per~~ report showing how much revenue a company made last month.

Source: McKinsey & Company, 2024

7. Diagnostic Analytics: This type of analytics tries to explain why something happened by digging deeper into the data.

Example: Finding out that sales dropped because of load shedding affecting store operations.

Source: IBM (2024)

8. Predictive Analytics: Uses past data to make predictions about the future. It often uses patterns and models.

Example: Banks' predicting which customers are likely to default on loans.

Source: SAS Institute (2024)

9. Prescriptive Analytics: This type of analytics suggests what actions a business should take based on the data. It goes beyond prediction to give recommendations.

Example: A delivery company suggesting the best routes to avoid traffic in ~~Joh~~ Johannesburg CBD.

Source: Deloitte (2024)

10. Data Literacy: The ability to read, understand and use data correctly. ~~It means~~

Example: A store manager being able to understand a sales report and make decisions from it.

Source: Glatner, 2025

Section 2: Data, Measurement and Reporting

11. Structured Data: Data that is organised in rows and columns, like in Excel or databases. It is easy to store and analyse.

Example: Customer list with names, ages and phone numbers

Source: IBM (2024)

- 12 Unstructured data: Data that is not organised in a fixed format, like text, videos or images. It is much harder to analyse.

Example: Comments on a X post about a brand.

Source: Oracle

- 13 Data Warehouse: A central place where a company stores large amounts of data for analysis and reporting.

Example: A bank storing all transactions data in one system for reporting.

Source: Amazon AWS

- 14 Data Mining: The process of finding patterns or trends in large sets of data.

Example: Discovering that customers who buy bread also often buy milk.

Source: IBM (2024)

15. Data Visualisation: Presenting data using charts, graphs or visuals to make it easier to understand

Example: A bar chart showing monthly sales

Source: Tableau

16. Dashboard: A visual display that shows important data and KPIs in one place, usually updated in real time

Example: A sales dashboard showing daily performance of stores.

Source: Microsoft

17. Metric: Any number used to measure something in a business

Example: Number of website visits

Source: HubSpot

18. Key Performance Indicator (KPI): A metric that is important for tracking business success and guiding decisions

Example: Monthly revenue target

Source: Investopedia.

19. Benchmark: A standard or reference point used to compare performance.

Example: Comparing your company's sales to industry averages.

Source: McKinsey & Company (2024)

20. Segmentation: Dividing customers into groups based on shared characteristics.

Example: Grouping customers by age, location or spending habits.

Source: Hubspot

Section 3: Customer & Sales KPI's

21. Conversion Rate: The percentage of people who take a desired action out of the total number of visitors.

Example: 10 out of 100 people buying a product = 10% conversion rate.

Source: Google Analytics

22. Customer Acquisition Cost (CAC): The total cost of getting one new customer, including marketing and sales expenses.

Example: Spending R1000 on ads to get 10 customers = R100 per customer

Source: Investopedia

23. Customer Lifetime Value (CLV): The total amount of money a business expects to earn from a customer over time.

24. Example: A customer who spends R500 every month for a year.

Source: Hubspot

24. Churn Rate: The percentage of customers who stop using a service over a period of time.

Example: People cancelling their DSTV Subscription

Source: Investopedia

25. Customer Retention Rate: The percentage of customers a company keeps over time. It is the opposite of churn.

Example: A gym keeping most of its members year after year.

Source: Hubspot

26. Net Promoter Score (NPS): Measures how likely customers are to recommend a business to others.

Example: Rating a service from 1 to 10 after using Uber

Source: Bain & Company

27. Average Order Value (AOV): The average amount of money spent per transaction

Example: Customers spending R250 per order at Takealot

Source: Shopify

28. Sales Funnel: Shows the journey customers take from first hearing about a product to making a purchase.

Example: Seeing an Instagram ad then visiting the website

Source: Hubspot

Section 4: Marketing and Campaign Analytics

29. Marketing Campaign: Planned set of activities to promote a product or service.

Example: A Black Friday sale by Checkers

Source: Hubspot

30. Post-campaign analysis: Reviewing the results of a campaign to see what worked and what didn't

Example: Checking if a promo increased sales after it ended.

Source: Google Analytics

31. Incremental Revenue: Extra income generated directly from a campaign or action.

Example: Extra sales made during a promotion period.

Source: Investopedia

32. Return on Investment (ROI): Measures how much profit is made compared to the cost of an investment.

Example: Spending R1000 and making R2000 profit.

Source: Investopedia

33. Return of Ad Spend (ROAS): Measures how much revenue is earned for every Rand spent on advertising.

Example: Spending R500 on ads and making R2000 revenue.

Source: Google Ads

34. Click-Through Rate (CTR): The percentage of people who click on an ad after seeing it.

Example: 5 clicks out of 100 views = 5% CTR.

Source: Google Ads.

35. Bounce Rate: Percentage of people who visit a website and leave without interacting.

Example: Someone opening a page and leave immediately.

Source: Google Analytics

26. A/B Testing: Compares two versions of something to see which performs better.

Example: Testing two different ad designs

Source: Optimizely

27. Lead Generation: Process of attracting and capturing potential customers.

Example: Getting people to sign up for a newsletter

Source: HubSpot

28. Marketing Attribution: Identifies which marketing channel contributed to a sale or result.

Example: Knowing whether a sale came from Instagram, Google or email.

Source: Google Analytics